

Weierstrass theorem and intermediate value theorem



Outline

Weierstrass theorem

Intermediate value theorem

Existence theorem for the n -th root

Weierstrass theorem

Introduction: Weierstrass theorem

- ▶ In this section we study when a continuous function has a **maximum** and a **minimum**
- ▶ The Weierstrass theorem guarantees the existence of maximum and minimum for continuous functions on closed and bounded intervals
- ▶ The proof uses a **bisection** construction, similar to that of the intermediate zero theorem
- ▶ Finally, through counterexamples, we will see that all the hypotheses of the theorem are essential

- ▶ The following theorem establishes **sufficient but not necessary** conditions for a function to have a maximum and a minimum.

Theorem 1: Weierstrass theorem

If a function $f : [a, b] \rightarrow \mathbb{R}$ is continuous on the interval $[a, b]$, then it has a maximum M and a minimum m in $[a, b]$.

Maximum and minimum points (1/2)

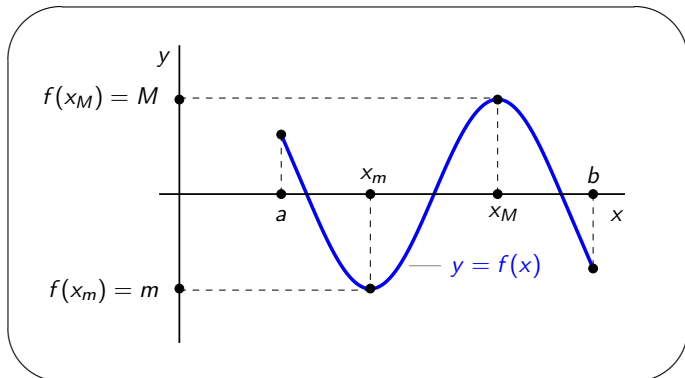
Under the hypotheses of the theorem there exist:

x_m and $x_M \in [a, b]$ such that

$f(x_m) \leq f(x) \leq f(x_M)$ for every $x \in [a, b]$

- ▶ We say that x_m is a **minimum point** of f and $m = f(x_m)$ is the **minimum** of f .
- ▶ We say that x_M is a **maximum point** of f and $M = f(x_M)$ is the **maximum** of f .

Maximum and minimum points (2/2)



Properties of the supremum/infimum

- ▶ Given two non-empty subsets E_1, E_2 of $\mathbb{R}$ we have:

$$\sup (E_1 \cup E_2) = \max (\sup E_1, \sup E_2)$$

- ▶ This property holds for both bounded and unbounded sets. If one or both sets are unbounded above we have:
 $\sup(E_1 \cup E_2) = +\infty$.

- ▶ For the infimum we have:

$$\inf (E_1 \cup E_2) = \min (\inf E_1, \inf E_2)$$

Proof (1/6)

We prove that f has a maximum in $[a, b]$. Consider the function:

$$f : [a, b] \rightarrow \mathbb{R}, \quad \text{and set} \quad \ell = \sup_{[a, b]} f$$

We now need to prove that:

- i) ℓ is finite, that is, $\ell \in \mathbb{R}$;
- ii) ℓ is equal to $f(x_0)$ for some $x_0 \in [a, b]$ (and hence ℓ is the maximum and x_0 is the maximum point).

Proof (2/6)

We split the interval $[a, b]$ into two equal intervals: I_1 and I_2 ; by the properties of the supremum we have:

$$\sup_{[a,b]} f = \max \left(\sup_{I_1} f, \sup_{I_2} f \right)$$

that is

$$\sup_{[a,b]} f = \sup_{I_1} f \quad \text{or} \quad \sup_{[a,b]} f = \sup_{I_2} f$$

Hence for one of the two intervals, which we call $[a_1, b_1]$, it will be true that:

$$\ell = \sup_{[a_1, b_1]} f$$

We now split $[a_1, b_1]$ into two equal intervals; for one of them, which we call $[a_2, b_2]$, it will be true that:

$$\ell = \sup_{[a_2, b_2]} f$$

Proof (3/6)

Proceeding by bisection in this way, we construct a sequence of intervals $[a_n, b_n]$, each contained in the previous ones, with the properties:

- 1) the sequence $\{a_n\}$ is monotone increasing and bounded, and the sequence $\{b_n\}$ is monotone decreasing and bounded;
- 2) $b_n - a_n = \frac{b-a}{2^n} \rightarrow 0$ as $n \rightarrow +\infty$;
- 3) $\ell = \sup_{[a_n, b_n]} f$ (the supremum of f is attained in the interval $[a_n, b_n]$)

By the same reasoning used in the proof of the intermediate zero theorem (which uses the monotonicity theorem for sequences), from points 1) and 2) it follows that the sequences $\{a_n\}$ and $\{b_n\}$ converge to the same limit $x_0 \in [a, b]$:

$$a_n \rightarrow x_0 \text{ and } b_n \rightarrow x_0 \text{ as } n \rightarrow +\infty.$$

We now proceed by cases.

Proof (4/6)

Case 1: $\ell \in \mathbb{R}$

If $\ell \in \mathbb{R}$, for every n there exists a point $t_n \in [a_n, b_n]$ such that:

$$\ell - \frac{1}{n} < f(t_n) \leq \ell \quad (1)$$

Indeed, since $\ell - \frac{1}{n}$ is less than ℓ , which is the least upper bound of the values of $f(x)$ in $[a_n, b_n]$, $\ell - \frac{1}{n}$ is not an upper bound; hence there exists $t_n \in [a_n, b_n]$ with property (1).

Since $t_n \in [a_n, b_n]$, and we have:

$$a_n \rightarrow x_0 \quad \text{and} \quad b_n \rightarrow x_0 \quad \text{as} \quad n \rightarrow +\infty$$

$$\text{then} \quad t_n \rightarrow x_0 \quad \text{as} \quad n \rightarrow +\infty$$

by the comparison (squeeze) theorem.

Proof (5/6)

Again by the comparison theorem, (1) then gives

$$\lim_{n \rightarrow +\infty} f(t_n) = \ell$$

On the other hand, since f is continuous and $t_n \rightarrow x_0$, we have:

$$\lim_{n \rightarrow +\infty} f(t_n) = f(x_0) \quad \text{therefore} \quad f(x_0) = \ell.$$

Then, since ℓ is the sup of the values of f in $[a, b]$, ℓ is the **maximum** of f in $[a, b]$, and it is attained at the point x_0 , the **maximum point**. Therefore, in this case the theorem is proved.

Proof (6/6)

Case 2: $\ell = +\infty$

If $\ell = +\infty$, for every n there exists a point $t_n \in [a_n, b_n]$ such that:

$$f(t_n) \geq n \quad (2)$$

Reasoning as above, one proves that $t_n \rightarrow x_0$ for some $x_0 \in [a, b]$. Since f is continuous at x_0 :

$$\lim_{n \rightarrow +\infty} f(t_n) = f(x_0)$$

but by (2) we have

$$\lim_{n \rightarrow +\infty} f(t_n) = +\infty$$

which is a **contradiction**, since at x_0 the function must have a finite value. Hence this case cannot occur.

Similarly, one proves that the function has a minimum in $[a, b]$. □

Nature of the proof

- ▶ The proof is a constructive existence proof (similar to that of the intermediate zero theorem)
- ▶ However, it cannot be used algorithmically since, after each bisection, it is not known a priori which interval the supremum or infimum is associated with.

The hypotheses of the Weierstrass theorem are all essential (1/3)

1. The interval must be closed

Consider as a counterexample:

$$f(x) = x \quad \text{with } x \in (0, 1)$$

- ▶ The function is continuous on a bounded, but not closed, interval.
- ▶ In this case, the function has neither a maximum nor a minimum (its supremum, 1, and its infimum, 0, are not attained by the function).

The hypotheses of the Weierstrass theorem are all essential (2/3)

2. The interval must be bounded

Consider as a counterexample:

$$f(x) = x \quad \text{with } x \in \mathbb{R}$$

- ▶ The function is continuous on an unbounded interval but has neither a maximum nor a minimum (it is not even bounded).

The hypotheses of the Weierstrass theorem are all essential (3/3)

3. The function must be continuous

Consider as a counterexample:

$$f(x) = \begin{cases} x & \text{for } x \in (0, 1) \\ \frac{1}{2} & \text{for } x = 0 \text{ and } x = 1 \end{cases}$$

- ▶ The function is defined on a closed and bounded interval $[0, 1]$ but it is not continuous.
- ▶ The function has neither a maximum nor a minimum (its supremum, 1, and its infimum, 0, are not attained by the function).

Summary: Weierstrass theorem

Statement:

- ▶ f continuous on $[a, b] \Rightarrow$ there exist $x_m, x_M \in [a, b]$ with $f(x_m) \leq f(x) \leq f(x_M)$ for every $x \in [a, b]$
- ▶ Sufficient but not necessary conditions

Proof:

- ▶ Bisection: intervals $[a_n, b_n]$ with $\ell = \sup_{[a_n, b_n]} f$, shrinking to x_0
- ▶ Case $\ell \in \mathbb{R}$: $f(x_0) = \ell$; case $\ell = +\infty$: contradiction

Essential hypotheses:

- ▶ Closed interval, bounded interval, continuous function

Intermediate value theorem

Introduction: Intermediate value theorem

- ▶ A continuous function on $[a, b]$ not only has a maximum and a minimum, but also takes **all the values** between them
- ▶ The proof combines the Weierstrass theorem with the intermediate zero theorem
- ▶ We will see that without continuity the property may fail, and that it holds thanks to the properties of the real numbers (it fails in $\mathbb{Q}$)
- ▶ We will summarize everything by saying that the image of $[a, b]$ under a continuous function is the interval $[m, M]$

Theorem 2: Intermediate value theorem

If a function $f : [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$, then it has a maximum M and a minimum m in $[a, b]$ and it takes all the values between m and M .

Intermediate value property

We are under the hypotheses of the Weierstrass theorem, hence we have:

x_m and $x_M \in [a, b]$ such that

$$f(x_m) \leq f(x) \leq f(x_M) \quad \text{for every } x \in [a, b]$$

The intermediate value theorem further tells us that:

$$\forall \lambda \in (m, M), \quad \text{there exists } x(\lambda) \in [x_m, x_M]$$

such that $f(x(\lambda)) = \lambda$

This property is called the **intermediate value property**.

Proof (1/2)

From the Weierstrass theorem we have a maximum point x_M and a minimum point x_m such that $f(x_M) = M$ (maximum) and $f(x_m) = m$ (minimum) in $[a, b]$. Let

$$m < \lambda < M$$

and consider the function

$$g(x) = f(x) - \lambda \quad \text{with } x \in [x_m, x_M]$$

which is continuous since $f(x)$ is continuous on $[a, b]$ and $x_m, x_M \in [a, b]$.

Proof (2/2)

Hence:

$$g(x_M) = f(x_M) - \lambda = M - \lambda > 0$$

$$g(x_m) = f(x_m) - \lambda = m - \lambda < 0$$

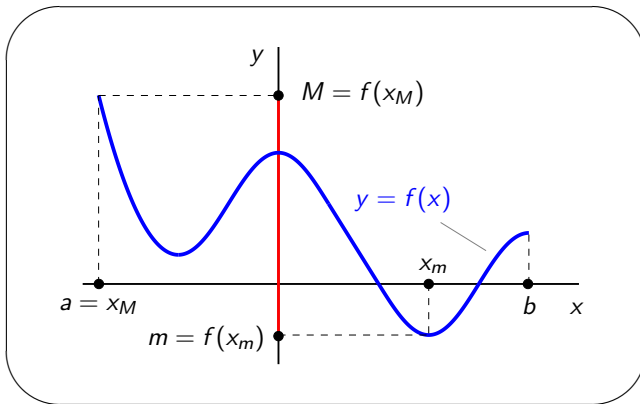
Then, by the intermediate zero theorem, there exists $x(\lambda) \in (x_m, x_M)$ such that:

$$g(x(\lambda)) = 0 \quad \text{that is} \quad f(x(\lambda)) = \lambda$$



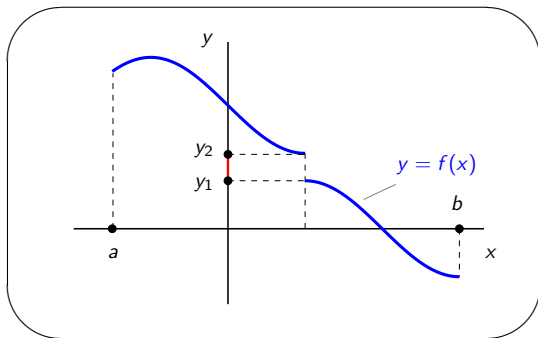
Graphical interpretation

- Given a function continuous on an interval $[a, b]$, graphically we have:



Example: discontinuous function without the intermediate value property

Consider for example the graph of the following function, discontinuous on $[a, b]$:



This function does not have the intermediate value property, that is, the values $\lambda \in (y_1, y_2)$ are not outputs of f .

- The properties of the two previous theorems can be summarized in the following single statement:

Corollary 1

If $f : [a, b] \rightarrow \mathbb{R}$ is continuous, then

$$f([a, b]) = [m, M];$$

i.e., the image of an interval $[a, b]$ is the interval with endpoints:

$$m = \min_{[a,b]} f \quad \text{and} \quad M = \max_{[a,b]} f.$$

Proof

It follows from the Weierstrass theorem and the intermediate value theorem. □

Example: failure of the intermediate value theorem in $\mathbb{Q}$ (1/2)

- ▶ Let

$$f(x) = x^2$$

and consider f as a function from the set $\mathbb{Q}$ of rational numbers to $\mathbb{Q}$ itself (this is legitimate because the square of a rational number is rational).

- ▶ Then f does not have the intermediate value property.
- ▶ Indeed, for example,

$$f(1) = 1, \quad f(2) = 4,$$

but f does not take all the rational values between 1 and 4: for example, it never takes the value 2, or 3.

Example: failure of the intermediate value theorem in $\mathbb{Q}$ (2/2)

- ▶ In other words, the intermediate value property holds for continuous functions **thanks to the properties of the set of real numbers**.
- ▶ This is a further reason why it is useful to work in the set of real numbers rather than in the set of rational numbers.

Summary: Intermediate value theorem

Statement:

- ▶ f continuous on $[a, b]$ takes all the values λ between m and M
- ▶ Proof: intermediate zero theorem applied to $g(x) = f(x) - \lambda$ on $[x_m, x_M]$

Corollary:

- ▶ $f([a, b]) = [m, M]$: the image of a closed and bounded interval is a closed and bounded interval

Limits of the property:

- ▶ It may fail for discontinuous functions
- ▶ It fails in $\mathbb{Q}$: the completeness of $\mathbb{R}$ is needed

Existence theorem for the n -th root

Introduction: Existence theorem for the n -th root

- ▶ As an application of the intermediate value theorem we prove that every positive real number has one and only one positive n -th root
- ▶ This result justifies the definition of $\sqrt[n]{y}$ for every $y > 0$
- ▶ Existence follows from the continuity of x^n , uniqueness from its strict monotonicity

Theorem 3

For every $y \in \mathbb{R}$, $y > 0$ and $n \in \mathbb{N}$, $n \geq 1$, there exists one and only one $x \in \mathbb{R}$, $x > 0$, such that $x^n = y$.

- This number x is called the **n -th root** of y .

Proof (1/3)

Consider the function $f(x) = x^n$.

This function is continuous on all of $\mathbb{R}$, because it is the product of n continuous functions $g(x) = x$.

We show that there exist

$$x_2 > x_1 > 0 \text{ such that } x_1^n < y < x_2^n.$$

Proof (2/3)

If $y = 1$ the n -th root exists and is unique; moreover:

1. If $y > 1$: it suffices to choose

$$x_1 = 1, x_2 = y \quad \text{and we have} \quad x_1^n < y < x_2^n \quad (1 < y < y^n)$$

2. If $y < 1$: it suffices to choose

$$x_1 = y, x_2 = 1 \quad \text{and we have} \quad x_1^n < y < x_2^n \quad (y^n < y < 1)$$

Proof (3/3)

Then we can apply the intermediate value theorem to the continuous function $f(x) = x^n$ on the interval $[x_1, x_2]$. Here its minimum is x_1^n and its maximum is x_2^n .

Since:

$$x_1^n < y < x_2^n, \text{ there exists } x_0 \in [x_1, x_2] \text{ such that } x_0^n = y.$$

Uniqueness follows from the fact that the function is strictly increasing. □

Summary: n -th root

Theorem:

- ▶ For every $y > 0$ and $n \geq 1$ there exists one and only one $x > 0$ with $x^n = y$

Proof:

- ▶ Find $0 < x_1 < x_2$ with $x_1^n < y < x_2^n$
- ▶ Existence: intermediate value theorem for x^n on $[x_1, x_2]$
- ▶ Uniqueness: x^n is strictly increasing

Summary (1/2)

Weierstrass theorem:

- ▶ A continuous function on $[a, b]$ has a maximum M and a minimum m
- ▶ Proof by bisection, using the properties of the supremum
- ▶ The hypotheses (closed interval, bounded interval, continuous f) are all essential

Intermediate value theorem:

- ▶ A continuous function on $[a, b]$ takes all the values between m and M
- ▶ It is proved with the intermediate zero theorem

Summary (2/2)

Corollary:

- ▶ If f is continuous on $[a, b]$, then $f([a, b]) = [m, M]$
- ▶ The intermediate value property depends on the properties of $\mathbb{R}$ (false in $\mathbb{Q}$)

n -th root:

- ▶ Every $y > 0$ has one and only one positive n -th root
- ▶ Existence from intermediate values, uniqueness from strict monotonicity